

Ethical Consideration When Design Traffic Flow Prediction System?

Phong Tran

104334842, Class 1-9 Thursday 8:30 - 10:30, John St, Hawthorn, Victoria, AU 3122

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I. EXECUTIVE SUMMARY

Nowadays, with the advancement in technologies, Traffic-based Route Guidance System plays a key role in navigating route and improve road distribution, therefore, minimizing the chance of traffic issues, such as congestion, accident. However, there are several concern that arise, such as route tracking related to data privacy, such as how the system collect user data to track the location on the real-time and giving the feedback easy and convenient. Secondly, several issues related to how AI implemented to the cost of computation, and cyber security threats such as adversarial attacks. Finally, ethical consideration existence surrounding on the algorithm of AI, with the rising of Natural Language Processing (NLP) combination with Large Language Models (LLMs), as a result, these black-box AI increases the potential risks. [1]

In this paper, we will discuss between our development with the time-frame given, some we finished some are not.

Making a connection between our previous work and the future work, what is need to be improved, and what is the issue when implementing that with advantage and disadvantage. Moreover, we will implement a privacy-focused with referring based on the several places on their law of privacy, such as Australia or EU. Furthermore, we need to make sure that the algorithm development need to be fair in terms of development. Finally, there needs to be transparency about the process and how in AI metrics, from training to the specific features given.

II. INTRODUCTION

Although we have completed our system integration, however, there are some arising issues that might affect on users and the system's performance. Moreover, to note on the key issue, ethics, which is the issues that rising among not only citizens but also government, to protect the secret data and protect for avoiding cyber attacks.

In this report, I will discuss about the search algorithm that I have been developed to integrate the program at the first stage with key analysis and comparison. Then giving the recommendations for further extension to test for a better and more suitable for all user's devices data. On the LSTM model, I only discuss on how the LSTM used by scratch comparison with using direct library and the ethical concern regarding to the processing data and training the LSTM model. Finally, our discussion on the Challenges in the Guidance System of Route based on Traffic.

III. FINDINGS

A. Search algorithm

1. Current Decision

Breadth-first search (BFS), uses traversal on graphs and trees, and rises on a single node then travel to particular node on nearby neighbors. It uses technique First In First Out, aka FIFO, utilizing the use of queue data structure. Its time complexity can be counted linearly with the edges number. [2]

Depth first search(DFS), also based similar to BFS, treats to the starting state on the problem when the root towards to a single path neighbors with given node. After that, the

ideal state or also goal state, located with the ideal branch. There is no need to go every level like BFS, and this stack data structure also known as Last In First Out (LIFO) [2]

Dijkstra algorithm stands in variants, first, uses greedy algorithm to ignore the obstacles distribution, traversing all nodes while searching at space. Moreover, it also can improve effectively on searching by start the search between start and end points simultaneously, minimizing the search space and cost of time. Finally, it improves substantially effective search to find the shortest path by using the information of landmarks and preprocessing, specifically for the path selection of complex environment like parking. [3]

2. Comparison

Based on the figure 1, We have been used for breadth and depth first search, both of them are really takes advantages for one is using for discovering broadly and in depth on the small branch, respectively. However, they takes longer time and speed than using Dijkstra, which only uses optimal paths.

Specifically, although we have seen the BFS, DFS, and Dijkstra are quite same each other in log time, but they are different in terms of path cost. To be more specific, DFS path is really high on all search algorithm, which might target to the dead-end, specifically in the real data of SCATS site data, which also known as nodes, and the edges is the connection among two SCATS with latitude and longitude. At the Dijkstra's algorithm, although it's high correctness and stability with addressing the single shortest path problem, it does not consider the negative weight edges. Due to the fact that it is not suitable for planning path in images with the edges of negative weight, which makes the path planning to fall. Moreover on the Dijkstra that it might require more space to store data, which will lead to larger requirement of memory space. [3]

3. Ethical implications

As mentioned above, the Dijkstra despite of higher accuracy than other search algorithm like BFS and DFS, it can take a lot of computation units, which is a challenge when user using their old devices to navigate route. Moreover, depending on the search algorithm, the CO2 will need to be consider when using these algorithm in order to save the fuel and minimizing travel distance as much as possible. Finally, it needs to be more transparent by letting users to ask whether why this search method for find the route is the best possible for them, like shortest by time or minimal as much as possible to avoid obstacle on the road.

4. Alternative

There are several consideration that we need to find out, for example, using the Genetic algorithm to wide taking for traffic route problems, such as Particle Swarm Optimization (PCO) and Ant Colony Optimization (ACO) because they are effectively reduce the chance of dangerous distance in path planning for multi-task, such as finding multiple path at the map. [4]

B. LSTM model implementation

1. Building from scratch versus using library available

When building from scratch, not only we gain deep understanding on the neural network model but also access to any components that hide Deep Learning (DL) inside, including five topics: (1) automatic differentiation (2) weight initialization (3) learning algorithms (4) regularization, and computations organization.

2. Train validate(val) test

In order to deal with the large volume of data, the train and val data need to be train and validate simultaneously each other before doing the test data. This will lower the bias among the training process and how it learned. Moreover, we need to make sure that there is a fair results and the framework of establishing accountability for the decisions of AI generated, by making the techniques become explainable of AI, defining clear boundaries and the framework of accountability establishing in order to make sure the complete project can be improved better and giving the ideas that can reshape the world in terms of fairness. [5]

3. Transparent how it be trained

As mentioned on Train validate (val) test, we can see that to make the AI traffic flow prediction be fair and accountable, make sure to define clear things. For example, all of the process in data processing, and training, specifically feature engineering, and model analysis need to be disclosed in order to given the fairness, reducing giving the bias result. Therefore, this also necessary to avoid legal procedure that cause disadvantages when there are something unfortunate happen, like traffic accident, congestion, or falling into the trap, which affects to the vehicle.

4. Alternative to consider

Comparing to GRU, it is better to choose it due to its removal of forget gates and fewer memory cell, which is sim-

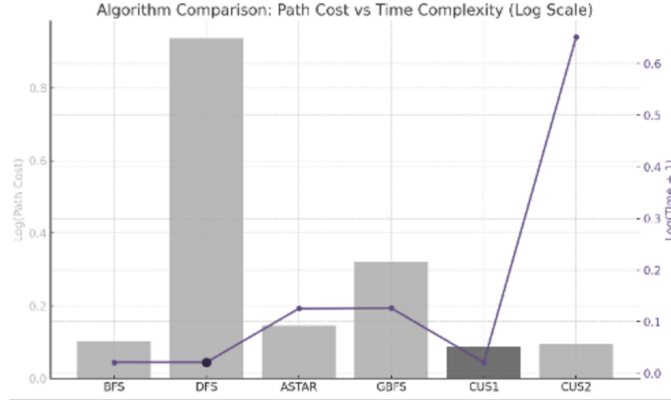


FIG. 1. Comparison path cost and time complexity in all five algorithm, especially BFS, DFS, and Dijkstra

plifier, and less computational cost to the user's devices.[6] Moreover, using Graph Neural Network has a combination of RNNs or attention, showing the road traffic has their graphs than grids. The Transformer from pure integrate with time series sequence to address the sequence issue. [7]

C. Challenges in the Guidance System of Route based on Traffic

In terms of security and privacy, there are some risks when implementing the System of Traffic-based Route Guidance. There are several things that we need to consider below, based on Divya,., 2024 et al [5]:

First of all, in terms of cyber security, the application needs to implement robust cyber methods, including prevent DDoS hacking, malware injection, penetration testing, and continuous monitoring tools. Also when it can be adversarial attacks that cause malware to affect the user's devices, the system needs to use security measures, and regular updates to fix bug and solve emerging threats.

Secondly, in terms of safety and reliability, the cyber functionalities at the system can be exploited for controlling malicious, therefore, utilizing strong measure, penetration testing, and secure protocols of communication. Then it need to be test on the real-world data, in order to get feedback for improvement, and to adapt the real-world on safety drivers.

Thirdly, it needs to follow the standard framework of legal law on the regulations of Australia and all over the world if expanding worldwide, which prioritize more on safety and innovation, and update the advancement on technology. Moreover, in order to mitigate the chance that accident involving the Guidance Route System of Traffic-based, we need to make sure that the framework is responsible to developer and operators.

Forth, in terms of acceptance and public trust, we need to ensure that increasing transparency in procedures testing, testing on the real-time with different people whom have an intendancy to survey, and prioritize safety in the best possible

mode in terms of permission.

Fifth, to solve the rare cases, we need to make sure that it needs to get feedback from users since they are understanding the most base on the problem. Moreover, the system needs to simulate extensively with make robust handle algorithm as we expected.

Sixth, at the ethical dilemmas, extensive data processing and make the training data diverse, continuous checking for bias, they are all necessary to prevent that model might be still existing bias if training data is bias. Moreover, it need to have the accessibility to make it suitable for diverse people, from young to old and people who are disabled, to avoid bias even in certain demographics, especially people who are disadvantaged.

Seventh, the user privacy is really serious, as we need to implement privacy techniques for preserving, informed consent for request, strictly followed on the regulation of data protection (GDPR) to protect the privacy of users.

Eighth, it always to monitor and encounter new bias or challenges on ethical issues as operating in the dynamic environments. In order to do that, we need to implement mechanism to monitoring constantly, looping for feedback, adapting model in different environment to consider and address rising ethical problems.

IV. CONCLUSION

To summarise, we have been talking about how search algorithm that currently developed, connecting with ethical issues among the users' device. Then the key issues on ethical consideration with the LSTM model with the processing and training data, as it needs to be transparent on the processing of using LSTM for predicting data with dataset given, which is public on VIC road. Finally, we make some key things, including: cyber security, safety and reliability, legal standard depending on countries, addressing on the rare cases, ethical dilemmas, user privacy, monitor and encounter new bias and ethical problem in the dynamic system.

There are several recommendation that I suppose that we need to do, first of all, it needs to be experimenting on other better search algorithm such as GBFS, A star, and Ant Colony Optimization (ACO), but also to make a robust system to find out which search algorithm is the best depending on user's system hardware availability. We also need to have a better data, with better guidance on training data and using various model such as GRU or Transformer to find that they are really better on the data training given to make a better predictions.

imize finding the relevant article and Consensus to find specific information to gain immediate understanding, Claude to structure my report with different heading and sub-headings with brief instruction. Google Scholar is still a vital knowledge search source to find the best and relevant article to deep analyze.

ACKNOWLEDGMENTS

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